E-Commerce Sales Data Analysis Using Python

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Simple student project for analysing online retail sales data.

[ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Load data

Keep `Online Retail.xlsx` in the same folder as this notebook.

[ ]:
```
df = pd.read_excel('Online Retail.xlsx')
df.head()
```

[ ]:
```
print(df.shape)
df.info()
```

2. Clean the data

[ ]:
```
df = df.drop_duplicates()
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
df['InvoiceNo'] = df['InvoiceNo'].astype(str)
df = df[~df['InvoiceNo'].str.startswith('C')]
df = df.dropna(subset=['CustomerID','Description'])
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]
df.head()
```

3. Create new columns

[ ]:
```
df['Revenue'] = df['Quantity'] * df['UnitPrice']
df['Year'] = df['InvoiceDate'].dt.year
df['Month'] = df['InvoiceDate'].dt.month
df['YearMonth'] = df['InvoiceDate'].dt.to_period('M').astype(str)
df['Day'] = df['InvoiceDate'].dt.day_name()
df['Hour'] = df['InvoiceDate'].dt.hour
```

4. Basic sales information

[ ]:
```
print('Total Revenue:', round(df['Revenue'].sum(),2))
print('Total Quantity:', df['Quantity'].sum())
print('Total Orders:', df['InvoiceNo'].nunique())
print('Total Products:', df['StockCode'].nunique())
print('Total Customers:', df['CustomerID'].nunique())
print('Average Order Value:', round(df['Revenue'].sum()/df['InvoiceNo'].nunique(),2))
```

5. Monthly revenue

[ ]:
```
monthly = df.groupby('YearMonth')['Revenue'].sum()
monthly.plot(kind='line', marker='o', figsize=(10,5))
plt.title('Monthly Revenue')
plt.xlabel('Month')
plt.ylabel('Revenue')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

6. Top countries

[ ]:
```
country = df.groupby('Country')['Revenue'].sum().sort_values(ascending=False).head(10)
print(country)
country.plot(kind='bar', figsize=(9,5))
plt.title('Top 10 Countries by Revenue')
plt.ylabel('Revenue')
plt.xticks(rotation=45)
```

```
plt.tight_layout()
plt.show()
```

7. Top products

[ ]:
```
products = df.groupby('Description')['Revenue'].sum().sort_values(ascending=False).head(10)
print(products)
products.sort_values().plot(kind='barh', figsize=(9,5))
plt.title('Top 10 Products by Revenue')
plt.xlabel('Revenue')
plt.tight_layout()
plt.show()
```

8. Top customers

[ ]:
```
customers = df.groupby('CustomerID')['Revenue'].sum().sort_values(ascending=False).head(10)
print(customers)
```

9. Conclusion

This analysis helps understand sales, products, customers and countries. Python was used for cleaning the data, calculating revenue and making simple charts.